Dictionary Coverage of Chinese Individual Classifiers

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1 Aim

This project asks how much classifier–target evidence in a Chinese corpus can be verified against two printed reference works—《现代汉语八百词》Xiàndài Hànyǔ Bābǎi Cí (Eight Hundred Words of Modern Chinese) and《汉语量词大词典》Hànyǔ Liàngcí Dà Cídiǎn (A Comprehensive Dictionary of Chinese Measure Words)—and what an automatic non-match actually means.

The initial plan was a direct dictionary lookup. Analysis showed, however, that an unmatched pair may reflect an annotation or target-span problem, tokenization, a polyfunctional classifier used in another grammatical role, or normal lexicographic selectivity rather than an invalid classifier–noun combination. The project therefore combines an explicit scope definition, automatic lookup over the core set, and focused manual review. It measures dictionary lookup coverage, not grammaticality.

2 Data and reference sources

The corpus is the LCMC and UCLA portion of Chinese Classifier Dataset v1.0 released with ClassifierGuesser (Peinelt et al. 2017). prepare_data.py merges the original train, dev and test files, keeps identifiers beginning with LCMC_ or UCLA_, and preserves the split label. The project is descriptive and does not train or evaluate a prediction model.

The prepared corpus contains 11,625 valid records (LCMC 6,001; UCLA 5,624), corresponding to 6,814 distinct classifier–target pair types.

Reference A is the classifier appendix of 《现代汉语八百词》Xiàndài Hànyǔ Bābǎi Cí (*Eight Hundred Words of Modern Chinese*), a noun-to-classifier list that yields exact pair evidence. Reference B is 刘子平 Liú Zǐpíng (2013), 《汉语量词大词典》Hànyǔ Liàngcí Dà Cídiǎn (*A Comprehensive Dictionary of Chinese Measure Words*), which is organized by classifier and provides example and entry-text evidence. Because neither source is distributed as structured data, both were converted locally into searchable files. The source scans and full extracts remain private pending distribution-rights checks.

3 Building the local reference lexicons

Three scripts in scripts/ prepare the dictionary evidence. lexicon.py parses the irregular plain-text transcription of Reference A, handling headings, glosses, full-width punctuation, and lines where the noun and classifier fields lost their separating whitespace. Unparseable lines raise an explicit error rather than being silently skipped.

dictionary_pdf.py extracts Reference B from a 280-page scanned PDF. Because the PDF has no structural entry boundaries, headings are recovered from typographic regularities: a headword is followed by a full-width pronunciation bracket, while implausibly long or punctuated candidates are rejected. Each recovered entry also stores its printed book page.

Extraction result	Count
Reference A noun–classifier pairs	799
Reference A distinct nouns / classifiers	368 / 135
Reference B entries recovered	1,057
Reference B entry text extracted	226,784 characters
Reference B printed pages covered	1–280

Table 1: Output of the two extraction scripts.

export_local_reference_lexicons.py runs both parsers and writes first.csv (noun, classifier) and second.jsonl (classifier, entry_text, book_pages). Extraction is separated from the main analysis because it requires pypdf and authorized source material, whereas the analysis itself uses only the Python standard library and the prepared files.

Reference B remains a candidate source because the extraction recovers entry blocks rather than internal structure: definitions, usage notes and examples are not separated reliably. A target string inside a block therefore does not by itself prove that the exact corpus relation is licensed.

4 Scope definition

The project targets lexically selective individual classifiers. Event and kind classifiers, container or standard measures, approximate measures, collective expressions, and ordinal/time expressions are outside the research question because their distribution is not determined by a noun-specific lexical pairing in the same way.

classifier_scope.json makes the decision auditable with three routes: include, exclude and review. scope.py validates the configuration, rejects duplicate category assignments, and sends any unlisted form to review rather than allowing it into the core experiment silently.

The productive individual classifiers 个 *gè* "general classifier", 位 *wèi* "polite classifier for people", 名 *míng* "classifier for people by name or role", and 员 *yuán* "classifier for personnel" are deliberately excluded because their wide distribution leaves little noun-specific lexical selection to measure. This removes 1,756 of the 6,814 pair types. Group-denoting forms such as 批 *pī* "batch", 群 *qún* "group", 套 *tào* "set", and 双 *shuāng* "pair" are also excluded because they quantify groups rather than individuating members.

The review group contains polyfunctional forms such as 片 *piàn* "slice or piece", 层 *céng* "layer", 块 *kuài* "piece or chunk", 道 *dào* "course or item", and 口 *kǒu*

Type in the taxonomy	Decision	Configuration category
Sortal › individual classifier	include	`specific_individual`
Sortal › individual, highly productive	exclude	`broad_productive`
Sortal › event classifier	exclude	`event_classifier`
Sortal › kind classifier	exclude	`kind_classifier`
Measure word › container	exclude	`container_measure`
Measure word › standard / approximate	exclude	`approximate_or_standard_measure`
Group-denoting expressions	exclude	`collective_classifier`
Ordinal, generation, time expressions	exclude	`ordinal_naming_or_time`
Polyfunctional, decided per pair	review	polyfunctional

Table 2: The scope configuration mapped onto the classifier taxonomy.

Scope decision	Pair types	Occurrences
Include – specific individual classifiers	2,001	3,613
Exclude – out-of-scope constructions	4,419	7,468
Review – polyfunctional forms	394	544
Total	6,814	11,625

Table 3: Scope of the 6,814 distinct pairs. Only included pairs form the denominator of the automatic core experiment.

"mouth or opening", whose function cannot be decided safely from the classifier string alone. The inventory is an operational research decision rather than a universal taxonomy.

5 Pipeline and modules

main.py is the project entry point. It parses the prepared records, applies the scope decision, performs dictionary lookup, integrates manual adjudication, and writes the generated outputs.

- prepare_data.py rebuilds private_data/corpus.csv from an authorized copy of the original dataset.
- scope.py loads and validates classifier_scope.json and classifies each classifier form.
- main.py extracts the target from <h> markup and the classifier from the <CL> marker; malformed records are recorded as invalid rather than silently dropped.
- Lookup uses a fixed priority: exact Reference A pair; Reference B example candidate; Reference B entry-text candidate; otherwise unmatched.
- adjudication.csv stores one manual label for each deferred review pair, which main.py routes back into the experiment.
- test_main.py contains eleven unit tests covering safe default-to-review behavior, invalid configurations and adjudication labels, malformed or duplicate input handling, and separation of automatic and adjudicated results.
- scripts/ contains the dictionary extraction pipeline described in Section 3.

Scope is decided before lookup. pairs.csv, scope_excluded.csv and scope_review.csv together cover the complete valid population, and their occurrence totals reconcile to 11,625; run_summary.json records the reconciliation. Non-core files also carry a diagnostic lookup status so that 'out of scope' is not confused with 'not found'.

Evidence status	Pair types	Occurrences
Exact Reference A pair	164	812
Reference B example candidate	118	329
Reference B entry-text candidate	231	562
Unmatched	1,488	1,910
Total	2,001	3,613

Table 4: Dictionary evidence for the automatic core experiment.

Reason the pair is unmatched	Sample	Share of the 1,488
Not a classifier–target relation at all	74	37.0% ± 6.2
Classifier not used as an individual classifier	12	6.0% ± 3.1
Valid use, documented by neither reference	111	55.5% ± 6.4
Context insufficient to decide	3	1.5% ± 1.6
Total	200	100%

Table 5: Manual analysis of a random sample of unmatched core pairs.

I also tested Stanza (Qi et al. 2020) as an exploratory method for recovering syntactic relations and head nouns. It introduced additional segmentation and dependency errors, especially in compounds and target-span mismatches, so it was retained only as a diagnostic experiment and not as part of the reproducible final pipeline.

6 Automatic dictionary coverage

Reference A provides exact evidence for 8.20% of core pair types. Adding Reference B candidates yields evidence for 513 of 2,001 types (25.64%) and 1,703 of 3,613 occurrences (47.14%). The higher token coverage shows that matched pairs are disproportionately frequent. Applying the same lookup to all 6,814 pairs without the scope filter gives 16.86% type coverage; this is only a baseline because it mixes in constructions deliberately outside the research question.

7 Why most core pairs are unmatched

I drew a fixed-seed simple random sample of 200 of the 1,488 unmatched core pair types and labelled each one manually with a single fine-grained category, first deciding whether the extracted pair is a classifier–target relation at all and, if so, why neither reference documents it. The ten working categories are aggregated into the four groups below. The percentages estimate the full unmatched population with 95% finite-population confidence intervals.

The first group consists of extraction or annotation artefacts. In 43 cases the target span is wrong; in 28 cases the marked classifier does not govern the extracted target; and three cases reflect tokenization or lexicalization errors. These cases cannot be fixed by better dictionary coverage. Representative malformed spans, corrected targets, and tokenization contrasts are presented with full interlinear information in Appendix A. The second group consists of genuine constructions in which the form has a non-individual reading: nine approximate-measure uses and three event-classifier uses.

Why the references do not document a valid pair	Sample	Share of the group
Compound whose head noun is documented	56	50.5% ± 8.7
Member of a documented semantic class	51	45.9% ± 8.6
Proper noun or loanword	4	3.6% ± 3.2
Total	111	100%

Table 6: Why the two references do not document a valid pair.

Route and manual labels	Pair types	Occurrences
Include – valid_combination	128	188
Exclude – approximate, event, container measure	112	153
Drop – not a classifier–target relation	145	192
Hold – grammatically wrong or uncertain	9	11
Total	394	544

Table 7: Manual adjudication of the deferred review scope.

Both groups follow from an editorial principle that Reference A states openly: nouns are collected 以类相从 *yǐ lèi xiāng cóng* "grouped by class". The appendix implements this principle in two ways. A corpus compound may inherit the classifier of its documented head noun, or an undocumented noun may belong to a semantic class represented by another member. These mechanisms correspond to the first two rows of Table 6; Appendix A provides the paired Chinese forms, pinyin, and English glosses.

Reference B encodes the same principle in its entry structure. Each sense begins with an intensional statement of the form 用于 X 等 *yòngyú X děng* "used for X and the like" and only then supplies examples. All 51 semantically extended types in the sample fall under such a statement, although string lookup can compare only against the printed examples. Appendix A separates the entry statements, loose boundary cases, and source-specific contrasts so that the linguistic evidence does not interrupt the main argument.

The lookup therefore compares one extension against another, whereas both references assert an intension and say so. Under this labelling scheme every valid undocumented case is attributable to one of the two routes above or to a proper noun or loanword; no sampled type was an ordinary simplex noun that the references could reasonably have listed but did not. The unmatched rate therefore reflects extraction noise and a documented difference between string-level lookup and lexicographic practice, rather than a set of genuine dictionary omissions.

The analysis is limited by type-based sampling, sparse coverage per classifier, and a single annotator with no second pass.

8 Manual adjudication of the review scope

The 394 review pair types (544 occurrences) are deferred because forms such as 片 *piàn* "slice or piece", 层 *céng* "layer", 块 *kuài* "piece or chunk", 道 *dào* "course or item", and 口 *kǒu* "mouth or opening" are polyfunctional. I manually labelled every deferred pair and routed it to include, exclude, drop or hold. The label concerns the linguistic relation, not dictionary coverage.

The 128 included pairs are well-formed individual-classifier uses. The 112 excluded pairs are legitimate quantificational constructions outside the target relation. The 145 dropped pairs are wrong target spans, wrong classifier markers, non-nominal targets, or lexicalization/tokenization artefacts; for example, 一块坐大巴 *yíkuài zuò dàbā* "take the bus together" contains 一块 *yíkuài* "together" as an adverbial rather than a classifier phrase.

Nine pairs are held because they were judged unacceptable or could not be resolved confidently. Routing the 128 valid uses back into the experiment yields an adjudicated core of 2,129 pair types and 3,801 occurrences, with coverage of 26.07% by type and 46.88% by occurrence. The denominator changes, but the headline coverage shifts by less than half a percentage point.

9 Discussion

The automatic and manual analyses converge on the same conclusion. Polyfunctional forms require human interpretation because segmentation, head identification, lexicalization and semantic function must all be correct before a pair can be evaluated lexically. This also explains why a general dependency parser did not solve the problem: an error at any one stage can produce a plausible-looking but linguistically invalid pair.

The manual review should not be extrapolated to the whole corpus. The 394 deferred types were selected precisely because they are difficult polyfunctional forms, not sampled randomly from all 6,814 pairs. Its purpose is to make the deferred population accountable and test whether routing it changes the result.

Methodologically, dictionary matching is useful as a reproducible baseline, but a non-match depends on the representation being compared. A dictionary compresses productive patterns through simplex heads and representative examples—Reference A declares this as the 以类相从 *yǐ lèi xiāng cóng* "grouped by class" principle and Reference B builds it into every entry—whereas a corpus lookup operates over observed strings. A coverage percentage measured this way is a statement about two inventories of units, and therefore requires qualitative interpretation before it can support linguistic claims.

10 Conclusion

Within the automatic core, the two reference works support 25.64% of pair types and 47.14% of occurrences; after manual adjudication of deferred polyfunctional forms, the figures are 26.07% and 46.88%. The stability of these values shows that adjudication clarifies the denominator without substantially changing the overall lookup result.

More importantly, the experiment shows that *compound inheritance* and *class analogy* are the main sources of productivity in classifier–noun combination: they extend a documented pairing to nouns that neither reference lists. Under compound inheritance, a

compound takes the classifier of its documented head noun; under class analogy, a noun takes the classifier of a documented member of the same semantic class. About 37% of the sampled unmatched types are annotation or extraction errors rather than classifier–target relations. Once these are removed, more than half of the remaining types—107 of 126 (84.9%)—follow one of these two rules and for that reason cannot be matched directly against either reference. They account for 96% of the valid but undocumented uses; the remainder are proper nouns and loanwords. Both rules follow the classifier appendix of 《现代汉语八百词》(Reference A), which notes that a dictionary cannot list every classifier–noun combination exhaustively and therefore relies on exactly these two rules: compound inheritance and class analogy.

References

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A Glossed Chinese examples

This appendix follows the presentation strategy used in recent classifier-prediction work: the main discussion remains in English, while Chinese evidence is collected in a structured table. Every Chinese form is accompanied by pinyin and an English gloss or functional description. "CL" marks the classifier position.

Table 8: Chinese evidence cited in the analysis, with pinyin and English glosses.

Evidence type	**Chinese**	**Pinyin**	**English gloss or function**
Wrong target span	一封密	*yì fēng mì*	one CL confidential…
Corrected target	一封密函	*yì fēng mìhán*	one confidential letter
Wrong target span	一幅层次	*yì fú céngcì*	one CL hierarchy…
Corrected target	一幅山水画	*yì fú shānshuǐhuà*	one landscape painting
Tokenization artefact	一句话说	*yí jù huàshuō*	one utterance says…
Erroneous token	话说	*huàshuō*	it is said
Intended target	话	*huà*	utterance
Reference A principle	以类相从	*yǐ lèi xiāng cóng*	grouped by class
Class name	树	*shù*	tree
Same class	桃树	*táoshù*	peach tree
Same class	松树	*sōngshù*	pine tree
Listed representative	教室	*jiàoshì*	classroom
Analogical member	办公室	*bàngōngshì*	office
Analogical member	卧室	*wòshì*	bedroom
Listed representative	老虎	*lǎohǔ*	tiger
Analogical member	狮子	*shīzi*	lion
Analogical member	豹子	*bàozi*	leopard
Inference label	可以类推	*kěyǐ lèituī*	can be inferred by analogy
Compound / head	条道路 / 条路	*tiáo dàolù / tiáo lù*	CL road / CL road
Compound / head	部线装书 / 部书	*bù xiànzhuāngshū / bù shū*	CL thread-bound book / CL book
Semantic extension	株水稻 / 株稻子	*zhū shuǐdào / zhū dàozi*	CL rice plant / CL rice plant
Semantic extension	头虎 / 头牛	*tóu hǔ / tóu niú*	CL tiger / CL cow
Semantic extension	封邮件 / 封信	*fēng yóujiàn / fēng xìn*	CL mail item / CL letter
Semantic extension	条刀痕 / 条伤疤	*tiáo dāohén / tiáo shāngbā*	CL knife mark / CL scar
Entry template	用于 X 等	*yòngyú X děng*	used for X and the like
Classifier entry	株：用于树木花草等	*zhū: yòngyú shùmù huācǎo děng*	classifier for plants: used for trees, flowers, grasses, and the like
Classifier entry	具：用于棺木、尸体等	*jù: yòngyú guānmù, shītǐ děng*	classifier for coffins and corpses: used for coffins, corpses, and the like
Classifier entry	条：用于细长或长条形的东西	*tiáo: yòngyú xìcháng huò chángtiáoxíng de dōngxi*	classifier for long narrow objects: used for long, narrow, or strip-shaped things
Loose boundary case	只蝴蝶结	*zhī húdiéjié*	CL bow knot
Loose boundary case	粒草芥	*lì cǎojiè*	CL grass or mustard seed
Reference B only	部面的	*bù miàndī*	CL minivan taxi
Reference B only	桩纠纷	*zhuāng jiūfēn*	CL dispute
Reference B statement	用于机器、车辆等	*yòngyú jīqì, chēliàng děng*	used for machines, vehicles, and the like
Reference B statement	用于事	*yòngyú shì*	used for matters or events